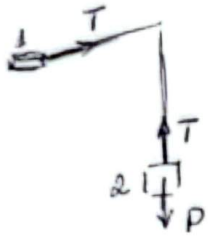
Colégio Agostiniano Mendel

Componente Curricular: Física – Dinâmica
Professor(a): Zuzana C. Páduca **Série:** 2ª **Data:** Junho-Agosto
Conteúdo: Força Centrípeta

1. a) $a_{cp} = \frac{v^2}{R} \Rightarrow a_{cp} = \frac{20^2}{200} \Rightarrow a_{cp} = 2\ m/s^2$; b) $F_{cp} = m\,a_{cp}$
$F_{cp} = 1200 \cdot 2 \Rightarrow F_{cp} = 2400\ N$
c) $F_{cp} = P - N$
$2400 = 12000 - N \Rightarrow N = 9600\ N$

2. $N - P = \frac{mv^2}{R} \Rightarrow N - 10000 = \frac{1000 \cdot 30^2}{100} \Rightarrow N = 19000\ N$

3. $T = \frac{mv^2}{R}\ (R = \ell) \Rightarrow 28{,}8 = \frac{0{,}1\,v^2}{0{,}5} \Rightarrow v^2 = 144 \Rightarrow v = 12\ m/s$

4. 

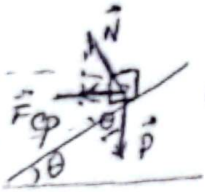
Bloco pendurado ⇒ repouso ⇒ $F_R = 0$
1 ⇒ $T = \frac{mv^2}{R}$
2 ⇒ $T = P \Rightarrow T = 10M$
$\frac{5}{1}$ ⇒ $M = \frac{2 \cdot 5^2}{1}$
$M = \frac{25}{5} \Rightarrow M = 5\ kg$

5. $P - N = \frac{mv^2}{R} \Rightarrow N = Mg - \frac{Mv^2}{R}$

6. 

$N + P = \frac{mv^2}{R} \Rightarrow N + 800 = \frac{80 \cdot 50^2}{50} \Rightarrow N = 4000 - 800 \Rightarrow N = 3200\ N$

7. $F_{at} = \frac{mv^2}{R} \Rightarrow \mu mg = \frac{mv^2}{R} \Rightarrow 0{,}8 \cdot 10 = \frac{v^2}{50} \Rightarrow v^2 = 400 \Rightarrow v = 20\ m/s$
$\sum F_{cp} = 0$ ($\vec{F}_{cp}$ ⊥ ao deslocamento)

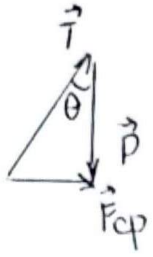
9. 

a) 

$\cos\theta = \frac{P}{N} \Rightarrow 0{,}6 = \frac{12000}{N}$
$N = 20000\ N$
b) $tg\theta = \frac{v^2}{Rg} \Rightarrow \frac{\operatorname{sen}\theta}{\cos\theta} = \frac{v^2}{Rg}$
$\frac{4}{3}\ \frac{0{,}8}{0{,}6} = \frac{v^2}{100 \cdot 10} \Rightarrow v^2 = \frac{4000}{3} \Rightarrow$
$v = \sqrt{1333} \Rightarrow v = 36{,}5\ m/s$

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10-



a) $F_R = F_{cp}$

$tg\theta = \frac{F_{cp}}{P}$

$\frac{sen\theta}{cos\theta} = \frac{F_{cp}}{P} \Rightarrow \frac{0,12}{0,4} = \frac{F_{cp}}{0,1} \Rightarrow F_{cp} = 0,3N$

b) $F_{cp} = \frac{mv^2}{R}$

$0,3 = \frac{0,1 \cdot v^2}{0,12}$

$v^2 = 0,36 \Rightarrow v = 0,6 m/s$

c) $v = \omega \cdot R \Rightarrow v = 2\pi f R \Rightarrow 0,6 = 2 \cdot 3 \cdot f \cdot 0,12$

$f = 0,83 Hz \quad (f \cong 0,8 Hz)$

11- * $T_A - P \cdot \cos 45° = \frac{mv^2}{R}$

$T_A - 400\frac{\sqrt{2}}{2} = \frac{40 \cdot 5^2}{2}$

$T_A - 400 \cdot 0,7 = 500$

$T_A = 500 + 280$

$T_A = 780N$

* $T_B - P_B = \frac{mv^2}{R}$

$T_B - 400 = \frac{40 \cdot 5^2}{2}$

$T_B = 500 + 400$

$T_B = 900N$

* $T_C - P \cdot \cos 30° = \frac{mv^2}{R}$

$T_C - 400 \cdot \frac{\sqrt{3}}{2} = \frac{40 \cdot 5^2}{2}$

$T_C - 400 \cdot \frac{1,7}{2} = 500$

$T_C = 500 + 340 \Rightarrow T_C = 840N$

12- $N = 3P \Rightarrow$ Sensação de aumento de peso $\Rightarrow$ posição + baixa

$N - P = \frac{mv^2}{R} \Rightarrow 3P - P = \frac{mv^2}{R} \Rightarrow 2P = \frac{mv^2}{R} \Rightarrow 2mg = \frac{mv^2}{R} \Rightarrow v = \sqrt{2Rg}$

13- a) * Iminência de cair $\Rightarrow N \to 0 \therefore v_{mín} = \sqrt{Rg}$

$v_{mín} = \sqrt{2 \cdot 10} = 2\sqrt{5} m/s$

b) * Sistema Conservativo

$E_{M1} = E_{M2}$

$mgh = \frac{mv_{m1n}^2}{2} + mg2R$

$10h = \frac{(\sqrt{20})^2}{2} + 10 \cdot 2 \cdot 2$

$10h = 10 + 40$

$h = \frac{50}{10}$

$h = 5m$

$E_{M1} = E_{M3}$

$mgh = \frac{mv_3^2}{2}$

$10 \cdot 5 = \frac{v_3^2}{2}$

$v_3^2 = 100$

$v_3 = 10 m/s$

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